

James Okooboh

Lagos, Nigeria

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Cloud & DevOps Engineer with hands-on experience architecting and deploying cloud-native infrastructure on AWS, including container orchestration with Kubernetes on Amazon EKS and ECS Fargate, GitOps continuous deployment with ArgoCD, and Infrastructure-as-Code automation with Terraform. Has shipped full-stack applications end-to-end, from Node.js and Java/Spring services through to production-style deployment pipelines backed by GitHub Actions CI/CD, Amazon RDS, and Prometheus/Grafana observability, including diagnosing and resolving live production incidents. Seeking Cloud Engineer, DevOps Engineer, Platform Engineer, or Site Reliability Engineer roles.

Core Competencies

- **Cloud Architecture (AWS):** EC2, S3, IAM, VPC, Subnets, Security Groups, Route 53, RDS, CloudWatch, Lambda, SNS, SQS, DynamoDB, API Gateway, Cognito, CloudFront, ACM, Secrets Manager
- **Container Orchestration & GitOps:** Docker, Kubernetes, Amazon EKS (Deployments, Services, Ingress, IRSA/OIDC), Amazon ECS on Fargate, ECR, ArgoCD (GitOps continuous deployment)
- **Infrastructure as Code:** Terraform (modularization, remote state, variable abstraction, multi-environment provisioning), Bash scripting
- **CI/CD Automation:** GitHub Actions - build, containerize, push to ECR/GitHub Packages, deploy
- **Monitoring & Observability:** Prometheus, Grafana, AWS CloudWatch (log aggregation, alerting, dashboards); production incident diagnosis and root-cause resolution
- **Cost Governance & Optimization:** AWS cost-aware infrastructure design and provisioning

Professional Experience

- **Miseacademy** **Remote**
DevOps & Cloud Engineering Intern *August 2026*
 - Designed and deployed production AWS infrastructure (EC2, RDS, S3, CloudFront, ACM, IAM, Secrets Manager) for a live e-commerce application serving real users on a custom domain
 - Automated multi-environment infrastructure provisioning with Terraform, including remote state management via S3 and DynamoDB locking
 - Containerized and orchestrated the application with Docker and Kubernetes, implementing GitOps continuous deployment with ArgoCD
 - Built a monitoring and alerting stack combining AWS CloudWatch, Prometheus, and Grafana across infrastructure and application layers
 - Diagnosed and resolved live production incidents, including cross-origin request failures, silent upload timeouts, and a false-positive alerting bug, tracing each to root cause rather than patching symptoms
- **Independent Training Program** **Lagos, Nigeria**
Cloud Engineering Instructor *2025–Present*
 - Designed and delivered a structured AWS and Linux cloud engineering curriculum for a cohort of 5 students, covering IAM policy design, EC2, S3, VPC networking, RDS, Route 53, and CloudWatch
 - Built and guided students through real-world cloud deployment labs on live AWS environments, taking students from zero to functional cloud deployments within the program timeline
 - Simplified complex networking and security concepts (VPC peering, security group chaining, NACL rules) through visual walkthroughs
 - Mentored students on Terraform/IaC best practices, including modularization, remote state, and variable abstraction

Key Projects

- *E-Commerce Platform on AWS with GitOps Deployment*
 AWS (EC2, RDS, S3, CloudFront, ACM, IAM, Secrets Manager), Docker, Kubernetes, ArgoCD (GitOps), Terraform, CloudWatch, Prometheus, Grafana · github.com/Jamesokooboh/e-commerce
 - Designed and deployed production AWS infrastructure (EC2, RDS, S3, CloudFront, ACM/SSL, IAM, Secrets Manager, Route 53) for a live e-commerce application running on a custom domain with real users
 - Automated infrastructure provisioning across 5 environments with Terraform, including remote state management via S3 and DynamoDB locking
 - Implemented GitOps continuous deployment with ArgoCD on a Kubernetes cluster, decoupling application deployment from the CI pipeline
 - Built a monitoring and alerting stack combining AWS CloudWatch, Prometheus, and Grafana across infrastructure and application-level metrics
 - Diagnosed and resolved live production incidents, including cross-origin request failures, a silent upload timeout, and a false-positive alerting bug, tracing each to root cause
- *Job Application Tracker*
 AWS, Terraform, Docker, GitHub Actions · github.com/Jamesokooboh/Job-Tracker
 - Designed and deployed a FastAPI + PostgreSQL service to AWS ECS Fargate using Terraform, provisioning VPC, ALB, RDS, IAM, and CloudWatch alarms from scratch with no manual console changes
 - Built a GitHub Actions CI/CD pipeline using OIDC federation for AWS auth (zero long-lived credentials), gated by automated tests and a manual production-approval step before infrastructure changes apply
 - Diagnosed and fixed real deployment failures – an IAM trust-policy mismatch against GitHub's OIDC token claims, an IAM least-privilege gap blocking role creation, and a CI/CD bootstrap ordering bug – verified via a live end-to-end deploy, then tore down cleanly
 - Locked down the network path (no public IPs on app/DB, security groups chained ALB→ECS→RDS) and kept secrets out of the repo entirely via AWS Secrets Manager
- *Full-Stack Web Application Deployment on Amazon EKS*
 Node.js, MySQL (RDS), Docker, Kubernetes (EKS), Terraform, GitHub Actions, Prometheus, Grafana · github.com/Jamesokooboh/restaurant-app-full-deployment-
 - Built and deployed a full-stack Node.js restaurant application to Amazon EKS with Kubernetes Deployments, Services, Ingress, and Secrets
 - Automated the full release process with GitHub Actions: build, push to ECR, update kubeconfig, apply manifests, roll out on every merge
 - Provisioned all supporting infrastructure with Terraform (custom VPC, EKS cluster, ECR repo, RDS MySQL)
 - Instrumented /health and /metrics endpoints scraped by Prometheus and visualized in Grafana
- *Agentic Terraform PR Reviewer – micro1 Frontier Engineering Challenge 2026*
 Python, Amazon Bedrock (Claude), Terraform, GitHub Actions, Checkov · github.com/Jamesokooboh/blast-radius
 - Built and benchmarked an AI agent reviewing Terraform pull requests via Amazon Bedrock, testing six agentic configurations against a one-shot prompt baseline across 15 labeled test cases
 - Found that added complexity – Terraform plan context, multi-agent orchestration, tool access – measurably hurt accuracy versus the simplest one-shot prompt (F1 0.784 down to 0.588–0.674), then shipped the winner as a GitHub Actions workflow scoped to least-privilege IAM
 - Caught a methodological error in the project's own early results by re-running the baseline three times, invalidating four of six initial “improvement” claims before publishing rather than after
- *Serverless Event-Driven Architecture*
 AWS Lambda, API Gateway, Cognito, S3, SQS, DynamoDB, SNS, Terraform, GitHub Actions · github.com/Jamesokooboh/serverless-infrastructure
 - Designed a fully event-driven serverless platform: API Gateway secured with Cognito triggers Lambda for synchronous requests
 - Built a decoupled, fault-tolerant async pipeline: S3 upload → SQS (with DLQ) → Lambda → DynamoDB → SNS notification

Certifications

- **DevOps & Cloud Engineering Internship Certificate** – Miseacademy, August 2026
- **micro1 Frontier Engineering Challenge 2026** – Certificate of Participation, September 2026
- **micro1 AI Interview Certification** – Outstanding Performance, September 2026